United States Patent Application Publication	20250284000
Kind Code	A1
Publication Date	September 11, 2025
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Method for Operating a Measuring Device

Abstract

A computer-implemented method for operating a measuring device, in particular a wall diagnostic device, includes (i) receiving radar data of a radar sensor unit of the measuring device, wherein the radar data depicts a wall to be diagnosed, (ii) performing wall diagnostics by performing an analysis of the radar data and providing diagnostic results via a diagnostic module of the measuring device, (iii) providing the diagnostic results and the uncertainty values via the diagnostic module on a display unit of the measuring device, and (iv) displaying the diagnostic results and the uncertainty values in the display unit.

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Family ID: **94820705**

Appl. No.: **19/073864**

Filed: **March 07, 2025**

Foreign Application Priority Data

DE	10 2024 202 240.3	Mar. 11, 2024

Publication Classification

Int. Cl.: **G01S13/88** (20060101); **G01S7/04** (20060101); **G01S7/41** (20060101); **G01S13/86** (20060101)

U.S. Cl.:

CPC **G01S13/888** (20130101); **G01S7/04** (20130101); **G01S7/417** (20130101); **G01S13/86** (20130101);

Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 240.3, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.
[0002] The present disclosure relates to a method for operating a measuring device, in particular a wall diagnostic device.
BACKGROUND
[0003] Diagnostic devices for diagnosing walls and detecting objects formed in the walls are known in the prior art.
[0004] It is an object of the present disclosure to provide an improved method of operating a measuring device, in particular a wall diagnostic device. It is further an object to provide an improved method for training an artificial intelligence of a measuring device.
[0005] The object is achieved by the method set forth below. Advantageous embodiments are subject-matter also set forth below.
SUMMARY
[0006] In one aspect, a method of operating a measuring device, in particular a wall diagnostic device is provided, wherein the method comprises: [0007] receiving radar data from a radar sensor unit of the measuring device, wherein the radar data depicts a wall to be diagnosed; [0008] performing a wall diagnostics by performing an analysis of the radar data and providing diagnostic results via a diagnostic module of the measuring device, wherein the wall diagnostic comprises: [0009] performing an object recognition of an object disposed in the wall using the diagnostic module, wherein the object recognition comprises an object detection and an object classification, and wherein the diagnostic results comprise at least one object position in the wall and/or an object type of the object; and [0010] determining uncertainty values for the diagnostic results by the diagnostic module, wherein the uncertainty values for the diagnostic results describe probability values that the diagnostic results match an actual state of the wall to be diagnosed, and comprise at least one probability value for the object position and/or a probability value for the object type of the object; [0011] providing the diagnostic results and the uncertainty values to a display unit of the measuring device using the diagnostic module; and [0012] displaying the diagnostic results and the uncertainty values in the display unit.
[0013] This can achieve the technical advantage that an improved method for operating a measuring device, in particular a wall diagnostic device, can be provided. To this end, radar data of a radar sensor unit of the measuring device which depicts the wall to be diagnosed is first received. Based on the received radar data, wall diagnostics are performed by a diagnostic module and diagnostic results are provided.
[0014] The wall diagnostics comprises performing an object recognition of an object disposed in the wall using the diagnostic module. The object recognition comprises object detection with determination of an object position and an object classification with determination of an object type of the object. The diagnostic results provided accordingly include at least the object position and the object type of the object disposed in the wall.
[0015] Further, uncertainty values for the diagnostic results are determined during the wall diagnostics. The uncertainty values here describe probability values for the indicated object position and probability values for the indicated object type. Further, the diagnostic results, including the uncertainty values, are provided by the diagnostic module to a display unit of the

measuring device and are displayed to a user in the display unit.
[0016] The method thus enables an object recognition of objects disposed in a wall to be diagnosed to be performed based solely on radar data of a radar sensor unit of the measuring device. In addition, uncertainty values for the object recognition, i.e. uncertainty values for the object position and the object type, are generated and displayed in the display unit. The uncertainty values can indicate a quality of the performed object recognition to the user. If there are high probability values for the object position or the object type, the user can assume that the object recognition is high quality, while low probability values indicate a low quality for the object recognition, respectively. In this way, the wall diagnostics performed by the measuring device can additionally be improved.
[0017] According to one embodiment, the wall diagnostics further comprises: [0018] performing a wall type classification and determining a wall type of the wall by the diagnostic module; and/or [0019] performing an object depth determination using the diagnostic module and determining an object depth of the object in the wall, wherein the object depth is defined as a distance of the object to a surface of the wall; and/or [0020] performing an object extension determination and determining an object extension of the object along a predefined direction via the diagnostics module, wherein the [0021] uncertainty values further comprise at least one probability value of the wall type and/or a probability value of the object depth and/or a probability value of the object extension of the object.
[0022] This has the technical advantage that a further improved wall diagnostics can be provided. For this purpose, the diagnostic module performs a wall type classification in the course of the wall diagnostics based on the radar data of the radar sensor unit and determines a wall type of the wall to be diagnosed. Alternatively or additionally, an object depth determination is performed and an object depth of the diagnostic module disposed in the wall is determined. The object depth describes a distance of the object to a surface of the wall. Alternatively or additionally, an object extension determination is performed based on the radar data and an object extension of the object is determined along a predefined direction.
[0023] Further, the uncertainty values comprise probability values for the wall type and/or probability values for the object depth and/or probability values for the object extension of the object. By determining the wall type and/or the object depth and/or the object type, the information provided by the wall diagnostics regarding the wall to be examined may be further increased. By taking into account the probability values and uncertainty values, the quality of the wall diagnostics may be determined and displayed in the display unit.
[0024] According to one embodiment, in the object recognition, a plurality of possible object positions and/or a plurality of possible object types of the object are determined as stand-alone diagnostic results, and/or wherein a plurality of possible wall types of the wall are determined as stand-alone diagnostic results in the wall type classification, and/or wherein a plurality of possible object depths are determined as stand-alone diagnostic results in the object depth determination and/or wherein a plurality of possible object depths of the object are determined as stand-alone diagnostic results in the object extension determination, wherein each of the plurality of diagnostic results is provided with a probability value, and wherein the plurality of diagnostic results and the plurality of probability values are displayed in the display unit.
[0025] This has the technical advantage that a further improved wall diagnostics can be provided. For this purpose, several alternatives of the determined diagnostic results, including corresponding uncertainty values, are provided in the wall diagnostics. Thus, multiple object positions, multiple object types, multiple object depths, multiple object extensions, and/or multiple wall types may be provided as results of the wall diagnostics. Corresponding uncertainty values are provided and displayed in the form of probability values for each of the plurality of object positions, plurality of object types, plurality of object depths, plurality of object extensions, and/or plurality of wall types.
[0026] The different probability values describe the certainty or uncertainty that the respective

object position, the respective object type, the respective object depth, the respective object extension and/or the respective wall type corresponds to the actual object position, the actual object type, the actual object depth, the actual object extension and/or the actual wall type. The user can thus be provided with additional information from the wall diagnostics.

[0027] According to one embodiment, the method furthermore comprises: [0028] providing a selection function, wherein at least one of the displayed diagnostic results can be selected by a user of the measuring device performing the selection function.

[0029] This has the technical advantage that the wall diagnostics can be further improved. A selection function is provided for this purpose to the user. By way of the selection function, the user can select one or more of the provided diagnostic results as the correct diagnostic result. The other diagnostic results, on the other hand, are not considered further in the wall diagnostics. For example, the user can select one of the specified wall types. For example, the user already knows the wall type of the wall to be examined. By selecting the corresponding wall type, it can be taken into account in the wall diagnostics accordingly, i.e., in the object detection. This may further improve the quality of the wall diagnostics.

[0030] According to one embodiment, the method furthermore comprises: [0031] receiving selection commands of the selection function, wherein one of the displayed diagnostic results is selected by the user using the selection commands; and [0032] providing feedback information to an external server unit, wherein the feedback information comprises the selection commands and the diagnostic results selected there along with the respective radar data.

[0033] This may achieve the technical advantage that the wall diagnostics can be improved by the provision of feedback information. For this purpose, when the selection function is activated by the user, the corresponding selections are registered and provided as feedback information to an external server unit. Based on the feedback information provided, the diagnostic module may be improved, for example, by re-training the diagnostic module. The correspondingly improved, i.e., newly trained diagnostic module can be installed on the measuring device in the form of an update in order to thus enable improved wall diagnostics.

[0034] The feedback information provided to the external server unit comprises the radar data on which the wall diagnostics were originally performed, the diagnostic results provided, and the diagnostic results selected by actuation of the selection function by the user. Based on this, the quality of the originally performed wall diagnostics may be determined by the external server unit. Based on this, the wall diagnostics, i.e. the diagnostic module, can be improved by re-training via the external server unit.

[0035] In the sense of the application, feedback information comprises any information that is based on an action taken by the user and that can be interpreted as feedback on the quality of the performed wall diagnostics. The feedback information can be direct, if necessary text-based feedback from the user. Alternatively or additionally,

[0036] According to one embodiment, the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor for providing additional sensor data, wherein the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.

[0037] The technical advantage can thereby be achieved that, through further sensor data, additional sensors, each of which is configured to detect different physical variables, can provide additional information in addition to the radar data of the radar sensor unit, for incorporation into the wall diagnostics. This additional information, which is preferably complementary to the information of the radar data of the radar sensor unit, allows further precise specification of the wall diagnostics or the object recognition.

[0038] According to one embodiment, the diagnostic module comprises at least one correspondingly trained artificial intelligence configured to perform an object recognition and/or wall classification and/or object depth determination based on the radar data and/or the additional sensor data

[0039] This may achieve the technical advantage that, because the diagnostic module is formed as a correspondingly trained artificial intelligence, that is trained based on the radar data, or, if applicable, taking into account the information of the additional sensors, to perform an object recognition and/or a wall classification and/or an object depth determination and/or an object extension determination, a reliable and powerful diagnostic module can be provided. By using the artificial intelligence technique, precise wall diagnostics can be provided.

[0040] According to one embodiment, the object classes of the object type of the object comprise: metal/non-metal object, low voltage cable, single-phase AC signal cable, multi-phase AC signal cable, wood beam, metal beam, plastic pipe, water-filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, and/or wherein the wall type classes of the wall type of the wall comprise: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, floor heating, wall heating.

[0041] This may achieve the technical advantage that objects and walls of different types may be detected and classified.

[0042] In one aspect, there a method for training an artificial intelligence of a wall diagnostic measuring device is provided, comprising: [0043] providing a training dataset for training an artificial intelligence, wherein the training dataset comprises a wall and a radar data depicting an object embodied in the wall as well as the feedback information provided according to the method of operating a measuring device; [0044] training the artificial intelligence based on the training dataset and taking into account the feedback information to perform object recognition of an object formed in a wall, wherein the object recognition comprises at least one object detection and an object classification.

[0045] In this way, the technical advantage can be achieved that improved training of the artificial intelligence of the wall diagnostic device, in particular post-training, is enabled, in which feedback is taken into account by the user of the wall diagnostic devices.

[0046] Provided according to one aspect is a computing unit configured to perform the method of operating a measuring device according to one of the preceding embodiments and/or the method of training an artificial intelligence.

[0047] According to one aspect, a computer program product is provided, comprising instructions that, when the program is executed by a data processing unit, cause the data processing unit to perform the method for operating a measuring device according to one embodiment and/or the method for training an artificial intelligence.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0048] Embodiments of the disclosure are described with reference to the following figures. The figures show:

[0049] FIG. **1** a schematic illustration of a measuring device according to one embodiment;

[0050] FIG. **2** a further schematic illustration of the measuring device according to a further embodiment;

[0051] FIG. **3** a further schematic illustration of the measuring device according to a further embodiment;

[0052] FIG. **4** a schematic illustration of a measurement of the measuring device according to one embodiment,

[0053] FIG. **5** a further schematic illustration of the measuring device according to a further embodiment;

[0054] FIG. **6** a schematic illustration of a system for operating a measuring device according to one embodiment,
[0055] FIG. **7** a flowchart of a method for operating a measuring device according to one embodiment of the disclosure,
[0056] FIG. **8** a further flowchart of the method for operating a measuring device according to a further embodiment,
[0057] FIG. **9** a flowchart of a method for training an artificial intelligence of a measuring device according to one embodiment, and
[0058] FIG. **10** a schematic representation of a computer program product.

DETAILED DESCRIPTION

[0059] FIG. **1** shows a schematic representation of a measuring device **100** according to one embodiment.

[0060] The present disclosure relates to a measuring device, in particular to a wall diagnostic device for examining walls **105** to be processed. Wall diagnostic devices are known in the prior art that are used to detect objects disposed in walls. Such devices allow the user to examine walls to be processed to search for objects disposed in the walls in order to be able to perform planned work, for example drilling in walls, based on this such that damage to the objects disposed in the walls can be avoided.

[0061] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by the user, the display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** to various operating modes.

[0062] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined and radar signals reflected by the wall **105** may be received.

[0063] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.

[0064] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform a corresponding diagnosis of the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** disposed within the wall **105**.

[0065] The wall diagnostics carried out by the diagnostic module **107** comprises at least performing an object recognition. The object recognition here comprises an object detection and an object classification of the object **113** disposed in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position here describes the positioning of the object disposed in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.

[0066] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** disposed in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to the user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.

[0067] The object **113** can be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.

[0068] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.

[0069] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.

[0070] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.

[0071] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105** and radar signals reflected from the wall **105** or the objects **113** disposed there are received via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.

[0072] For example, diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** disposed in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.

[0073] The diagnostic results **109** configured in this manner may subsequently be displayed to the user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.

[0074] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.

[0075] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than that given by the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0076] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device

100 is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.

[0077] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-described wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** disposed in the wall **105**. The object classification or determination the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.

[0078] The object classes may comprise: metal/non-metal object, low voltage cable, single-phase AC signal cable, multi-phase AC signal cable, wood beam, metal beam, plastic pipe, water-filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.

[0079] Further, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**. Possible wall types **123** may comprise: concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.

[0080] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.

[0081] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can in particular be used for object recognition of the objects **113** disposed in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.

[0082] In particular, for example, the material of the objects **113**, for example as a metallic or non-metallic material, can be improved and classified by using the additional sensor information.

[0083] FIG. **2** shows another schematic representation of the measuring device **100** according to a further embodiment.

[0084] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing of the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.

[0085] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall to be examined **105** and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.

[0086] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be included in the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0087] In the embodiment shown, the diagnostic module **107** comprises a wall type classification

module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. In the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.
[0088] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.
[0089] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects on the radar data **103** can occur.
[0090] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **133** performs the object recognition of the object **113** disposed in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.
[0091] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.
[0092] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.
[0093] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.
[0094] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, for example, comprising the wall type classification module **129** and/or the object recognition module **131** according to the embodiment in FIG. **2**, and a post-processing module **163**.
[0095] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information of the additional sensors may also serve as input data for the wall diagnostics. Here, the different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon

completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual sub-analysis results to form the diagnostic results **109** of the wall diagnostics.

[0096] Alternatively or additionally, different sub-aspects of the wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0097] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moved relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0098] The different processing paths perform a stand-alone wall diagnostics in this respect, comprising at least determining the object position **115** and/or the object type **117** of the object **113** disposed in the wall **105**.

[0099] The summary module **165** may summarize the sub-results provided in the individual processing paths **102** of the stand-alone wall diagnostics of the different regions of the wall **105** into a contiguous diagnostic result **109**. The contiguous diagnostic result here describes the wall diagnostics of a contiguous spatial area that was covered during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.

[0100] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object disposed in the wall **113** can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0101] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may respectively be carried out. In the summary module **165**, the various sub-results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0102] The diagnostic module **107** can be divided into different artificial intelligences **125**, as already shown in the embodiment in FIG. **2**. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0103] The respective modules may each be configured as stand-alone artificial intelligences **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known in the prior art.

[0104] FIG. **4** shows a schematic illustration of a measurement of the measuring device **100** according to one embodiment.

[0105] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Further, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate the corresponding sensor data may be sampled. Further, the radar data **103** or the additional sensor information **104** may be transformed to the respective required frequency range or time range. Methods known from the prior art can be used for this purpose.

[0106] Further, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0107] Graph a) of FIG. **4** shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.

[0108] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement of the measuring device **100** ends.

[0109] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.

[0110] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.

[0111] A further temporal window **167** or spatial window **167** can be provided as soon as one or more scan points are available.

[0112] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. **3** as well. According to the embodiment of FIG. **2**, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.

[0113] As stated above, wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by rule-based algorithms. A combination of artificial intelligence and a rule-based algorithm is also possible within a processing path **102** in the form of a parallel connection or concatenation.

[0114] The diagnostic results **109** of the wall diagnostics may be implemented as numeric values, vectors, or matrices. Further, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be given for the object detection. The same may apply to the position and/or depth determination, for which corresponding probability values can also be given.

[0115] If, in addition to the radar data **103**, the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by rule-based combinations.

[0116] In the post-processing of each processing path **102**, of the embodiment of FIG. **3**, multiple algorithm results based on multiple windows **167** may be summarized by the summary module

165. This summary can in particular be realized by majority formation, sum formation or also by multiplication of successive probability values.

[0117] Further, by clustering multiple results, for example, it is possible to detect which objects of multiple detected objects lying close to one another are the same object so that they are not incorrectly detected multiple times.

[0118] Likewise, it is possible to multiply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic sub-results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic sub-results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic sub-results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.

[0119] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Here, weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.

[0120] For example, the result of an object classification in which the object type **117** of an object disposed in the wall **105** is defined can be utilized to increase the weight of a wall type classification in which the wall type **123** of the respective wall **105** is determined in the post-processing at locations without objects **113**, because the respective radar data **103** at these locations are less influenced by reflections of the objects **113**.

[0121] FIG. **5** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0122] Graphs a) and b) of FIG. **5** show two different alternatives of a joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.

[0123] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107**, based on this.

[0124] In graph b), in contrast, the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.

[0125] In an additional concatenation layer **148**, the partial results of the partial diagnoses of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.

[0126] The correspondingly configured diagnostic module **107** is designed to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.

[0127] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known in the prior art.

[0128] FIG. **6** shows a schematic illustration of a system **600** for operating a measuring device **100** according to one embodiment.

[0129] In the embodiment shown, the system for operating a measuring device **100** comprises an external server unit **114** in addition to the measuring device **100**.

[0130] According to the disclosure, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. The radar data **103** here depicts the wall **105** to be examined, including the objects **113** disposed therein.

[0131] The wall diagnostics are performed by the diagnostic module **107** based on the radar data **103** and corresponding diagnostic results **109** are generated. The wall diagnostics comprise performing at least one of object recognition and object detection, with determination of an object position **115** and an object classification, with determination of the object type **117** of the objects **113** disposed in the wall **105**.

[0132] According to the present disclosure, the wall diagnostics further comprises determining uncertainty values **110** for the diagnostic results **109**. The uncertainty values here describe probability values that the determined diagnostic results **109** and an actual state of the wall **105** to be diagnosed will match. According to the disclosure, the uncertainty values **110** comprise at least probability values with respect to the determined object position **115** and probability values for the determined object type **117**.

[0133] According to the present disclosure, the determined diagnostic results **109**, including the uncertainty values **110**, are provided by the diagnostic module **107** of the display unit **111** and are shown in the display unit **111**.

[0134] In the graph shown, the diagnostic results **109** and the corresponding uncertainty values **110** are plotted graphically in the display unit **111**.

[0135] The user can thus interpret the wall diagnostics or the provided diagnostic results **109** accordingly, taking into account the uncertainty values **110**.

[0136] According to one embodiment, the wall diagnostics may further comprise the object depth **119** or the object extension **121** of the object **113** disposed in the wall **105**, in addition to the object position **115** and the object type **117**. Additionally, the wall diagnostics may comprise determining the wall type **123** of the wall **105**. The object depth **119**, the object extension **121** as well as the wall type **123** can be displayed as corresponding diagnostic results **109** including corresponding uncertainty values **110** in the display unit **111**.

[0137] According to one embodiment, the wall diagnostics comprises determining a plurality of possible object positions **115**, possible object types **117**, possible object depths **119**, possible object extensions **121**, and/or possible wall types **123**. The various possible object positions **115**, object types **117**, object depths **119**, object extensions **121**, and/or wall types **123** may be provided with corresponding uncertainty values **110** and displayed in the display unit **111**.

[0138] According to one embodiment, a selection function **116** is furthermore provided to the user. By way of the selection function **116**, the user can select various results of the displayed diagnostic results **109** by actuating them. For example, the user may select the correspondingly proposed wall type **123** using the selection function **116** if they know the actual wall type **123**.

[0139] The correspondingly selected wall type **113** is subsequently taken into account in the remaining wall diagnostics, while the non-selected proposed wall types **123** remain unconsidered. The same applies to the additional diagnostic results **109** displayed accordingly.

[0140] According to one embodiment, feedback information **112** is provided to the external server unit **114** based on the received selection commands of the selection function **116**, wherein the displayed diagnostic results **109** are selected accordingly with the selection commands by the user. The feedback information **112** comprises the selection commands and the diagnostic results **109** selected therein along with the respective radar data **103**.

[0141] For example, the feedback information **112** may comprise information regarding which diagnostic results **109** the user selected as correctly representing the state of the wall **105**. Further, the feedback information **112** may comprise information as to which diagnostic results **109** the users have not selected as correctly representative.

[0142] Based on the feedback information **112**, the external server unit **114** may improve the diagnostic module in terms of wall diagnostic quality by training the algorithm of the diagnostic module **107**.

[0143] According to one embodiment, the feedback information **112** is provided by a plurality of different measuring devices **100**. The measuring devices **100** can be used, for example, by a plurality of users in normal operation. For example, during operation of the measuring device, the corresponding feedback information **109** is initially stored on a corresponding storage unit of the measuring device **100** and provided to the external server unit **114**.

[0144] According to one embodiment, the external server unit **114** is further configured to generate a training dataset **143**, taking the feedback information into account. Based on the training dataset **143**, re-training of the diagnostic module **107** is enabled, with re-training taking the feedback information **112** into account. The correspondingly retrained diagnostic module **107** may subsequently be installed in new measuring devices **100** and/or installed in already existing measuring devices **100** as an update to the existing diagnostic module **107**. By considering the feedback information in the training of diagnostic module **107**, the performance of the diagnostic module **107** can be improved.

[0145] FIG. **7** shows a flowchart of a method **200** for operating a measuring device **100** according to one embodiment.

[0146] To operate the measuring device **100**, in a first method step **201**, radar data **103** of the radar sensor unit **101** of the measuring device **100** is first received.

[0147] In a further method step **203**, the wall diagnostics are performed by the diagnostic module **107** of the measuring device **100** based on the radar data **103**.

[0148] For this purpose, in a method step **205**, a wall detection of the object **113** disposed in the wall **105** is performed by the diagnostic module **107**. The object recognition comprises at least one object detection with determination of the object position **115** and an object classification with determination of the object type **117** of the object **113**.

[0149] In a further method step **207**, corresponding uncertainty values **110** are determined for the generated diagnostic results **109**. The uncertainty values describe probabilities of the diagnostic results **109** being consistent with the actual state of the wall **105** being diagnosed.

[0150] In a further method step **209**, the diagnostic results **109** and the uncertainty values **110** of the display unit **111** of the measuring device **100** are provided. In a further method step **211**, the diagnostic results **109** and the uncertainty values **110** are displayed in the display unit **111**.

[0151] FIG. **8** shows a further flowchart of the method **200** for operating a measuring device **100** according to a further embodiment.

[0152] The embodiment in FIG. **8** is based on the embodiment in FIG. **7** and comprises all the method steps described there.

[0153] In the embodiment shown, the wall diagnostics further comprises performing a wall type classification and determining a wall type **123** in a method step **213**. Further, in a method step **215**, an object depth determination is performed and an object depth **119** of the object **113** is determined.

[0154] Further, in a method step **217**, an object extension determination is performed and the object extension **121** of the object **113** is determined.

[0155] Moreover, in a method step **219**, the selection function **116** is provided.

[0156] In a method step **221**, the selection commands of the selection function **216** are received, wherein one of the displayed diagnostic results is selected by the selection commands.

[0157] In another method step **223**, feedback information **112** is provided to an external server unit **114**. The feedback information **112** comprises the selection commands and the diagnostic results **109** selected therein, including the respective radar data **103**.

[0158] According to one embodiment, the wall diagnostics may also be performed in consideration of additional sensor information **104**.

[0159] FIG. **9** shows a flowchart of a method **400** for generating a training dataset for training an artificial intelligence **125** of a measuring device **100** according to one embodiment.

[0160] To train the artificial intelligence **125** of the measurement device **100** for wall diagnostics, in one a method step **401** a training dataset **143** is first provided for training the artificial intelligence **125**. The training dataset **143** comprises radar data **103** that depicts the walls **105** to be diagnosed, including the objects **113** formed in the walls **105**. Further, the training dataset **123** comprises the feedback information **112** provided according to the method **200** described above for operating the measuring device **100**. The feedback information **112** is included in the training dataset **143** as additional information in addition to the provided radar data **103**.
[0161] The radar data **103** can be based on measurements performed solely to create a training dataset. Alternatively or additionally, the radar data **103** may have been incorporated by a plurality of users during operation of a plurality of measuring devices **100** and provided to create the training dataset **143** of the external server unit **116**.
[0162] In another method step **403**, the artificial intelligence training is performed based on the previously generated provided training dataset **143**, taking the feedback information into account. The artificial intelligence **125** is trained to perform object recognition of objects **113** formed in walls **105**.
[0163] FIG. **10** shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method **200** for operating a measuring device **100** and/or the method **400** for training an artificial intelligence **125**.
[0164] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can in this case be any desired storage medium known from the prior art.

Claims

1. A method of operating a measuring device, comprising: receiving radar data of a radar sensor unit of the measuring device, wherein the radar data depicts a wall to be diagnosed; performing wall diagnostics by performing an analysis of the radar data and providing diagnostic results with a diagnostic module of the measuring device, wherein the performing wall diagnostics comprises: performing an object recognition of an object disposed in the wall using the diagnostic module, wherein the object recognition comprises an object detection and an object classification, and wherein the diagnostic results comprise at least one object position in the wall and/or an object type of the object; and determining uncertainty values of the diagnostic results using the diagnostic module, wherein the uncertainty values of the diagnostic results describe the probability values that the diagnostic results match an actual state of the wall to be diagnosed, and comprise at least one probability value of the object position and/or one probability value of the object type of the object; providing the diagnostic results and the uncertainty values by the diagnostic module to a display unit of the measuring device; and displaying the diagnostic results and the uncertainty values in the display unit.
2. The method according to claim 1, wherein the wall diagnostics further comprises: performing wall type classification and determining a wall type of the wall using the diagnostic module; and/or performing an object depth determination and determining an object depth of the object in the wall using the diagnostic module, wherein the object depth is defined as a distance of the object to a surface of the wall; and/or performing an object extension determination and determining an object extension of the object along a predefined direction using the diagnostic module, wherein the uncertainty values further comprise at least one probability value of the wall type and/or a probability value of the object depth and/or a probability value of the object extension of the object.
3. The method according to claim 1, wherein a plurality of possible object positions and/or a plurality of possible object types of the object are determined in the object recognition as stand-alone diagnostic results, and/or wherein a plurality of possible wall types of the wall are determined

as stand-alone diagnostic results in the wall type classification, and/or wherein a plurality of possible object depths are determined as stand-alone diagnostic results in the object depth determination and/or wherein a plurality of possible object extensions of the object are determined in the object extension determination as stand-alone diagnostic results, wherein each of the plurality of diagnostic results is provided with a probability value, and wherein the plurality of diagnostic results and the plurality of probability values are displayed in the display unit.

4. The method according to claim 3, further comprising: providing a selection function, wherein when a user of the measuring device executes the selection function, at least one of the displayed diagnostic results is selectable.

5. The method according to claim 4, further comprising: receiving selection commands of the selection function, wherein one of the displayed diagnostic results is selected by the user using the selection commands; and providing feedback information to an external server unit, wherein the feedback information comprises the selection commands and the diagnostic results selected there along with the respective radar data.

6. The method according to claim 1, wherein the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor for providing additional sensor data, and wherein the diagnostic module is configured to perform the wall diagnosis by taking into account the additional sensor data.

7. The method according to claim 1, wherein the diagnostic module comprises at least one correspondingly trained artificial intelligence configured to perform object recognition and/or a wall classification and/or an object depth determination based on the radar data and/or the additional sensor data.

8. The method according to claim 1, wherein object classes of the object type of the object comprise: metal/non-metal object, low voltage cable, single-phase AC signal cable, multi-phase AC signal cable, wood beam, metal beam, plastic pipe, water-filled plastic pipe, non-water filled plastic pipe, and/or wherein the wall type classes of the wall type of the wall comprise: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, floor heating, wall heating.

9. A method for training an artificial intelligence of a measuring device used for wall diagnostics, comprising: providing a training dataset for training the artificial intelligence, wherein the training dataset comprises a wall and radar data depicting an object formed in the wall, as well as the feedback information provided according to claim 5; and training the artificial intelligence based on the training dataset and taking into account the feedback information for performing object recognition of an object formed in a wall, wherein the object recognition comprises at least one of object detection and object classification.

10. A computing unit configured to perform the method for operating a measuring device according to claim 1.

11. A computer program product comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method for operating a measuring device according to claim 1.

12. The method according to claim 1, wherein the measuring device is a wall diagnostic device.

13. The method according to claim 8, wherein: the water-filled plastic pipe is a fresh water pipe, and the non-water filled plastic pipe is a waste water pipe.

14. A computing unit configured to perform the method for training an artificial intelligence of a measuring device to perform wall diagnostics according to claim 9.

15. A computer program product comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method for training an artificial intelligence of a measuring device to perform wall diagnostics according to claim 9.